A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Nutrient pollution, sewage treatment & acid mine drainage - Set 1

Mark scheme

AQA spec 3.4.3.2.9-3.4.3.2.10; Pollution; textbook pp. 266-275

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Dissolved oxygen fluctuates daily / 24-hour cycle: highest in the afternoon (about 13-13.5 mg dm^{-3} at 16 and 40 h) and lowest at night / early morning (2.5-3.5 mg dm^{-3} at 4 and 28 h) **1** The night-time minimum is lower on the second night (2.5 vs 3.5) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 4 points below. **1** Nitrates (from fertiliser runoff) and phosphates (from sewage effluent) cause eutrophication: rapid growth of algae / cyanobacteria (algal bloom) **1** During the day photosynthesis by the dense algae releases oxygen so concentrations rise (above the normal saturation level) **1** At night there is no photosynthesis but respiration by algae / animals / bacteria continues, using oxygen **1** Aerobic bacterial decomposition of dead algae / shaded macrophytes uses oxygen – warm summer water holds less oxygen and decay is rapid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	4
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** At night no photosynthesis replaces the oxygen used in respiration / decomposition so concentrations fall below the minimum fish can tolerate **1** At the end of summer large amounts of dead vegetation decay rapidly in warm water causing severe deoxygenation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water	2
01.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Cyanobacteria release toxins harmful to livestock, pets and people **1** Floating algae shade submerged / rooted macrophytes which die, disrupting food webs **1** Algae are an unreliable food source (die off rapidly) so food webs are poor **1** Blocked filters / unpleasant taste and smell in water supplies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-268; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Bacteria in the gut convert nitrates to nitrites **1** Nitrites react with haemoglobin so it cannot carry oxygen **1** Infants are most at risk because they have more of the gut bacteria and their haemoglobin reacts faster – usually from shallow wells receiving arable drainage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrites may be converted in the gut to nitrosamines which are carcinogenic **1** But most research has been on rats / the evidence in humans is not conclusive – nitrites are also ingested in processed food so the source cannot be isolated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Reduce use of nitrate fertilisers / match application to crop need **1** Avoid ploughing / applying fertiliser at times of heavy rain (autumn / winter) **1** Use low-solubility / slow-release fertilisers rather than soluble rapid-release ones **1** Store manure / slurry where it cannot leach or wash into rivers **1** Grow crops with lower nitrogen requirements (eg triticale instead of wheat) **1** Grow legumes with nitrogen-fixing bacteria instead of relying on artificial fertiliser **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **nitrogen-fixing** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Iron(III) sulfate is added to effluent: phosphates are precipitated as insoluble iron phosphate (a fine sediment that can be used as fertiliser) **1 Lakes:** dredging phosphate-rich sediments / adding iron(III) sulfate to reduce phosphate solubility (eg the Norfolk Broads) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 267-268; AQA spec 3.4.3.2.9.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Screens (metal grills) trap floating / suspended non-biodegradable items (paper, plastic) which are landfilled or incinerated **1 Grit traps:** the channel widens so flow slows and stones / road grit sink out because there is not enough kinetic energy to carry them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers	2
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** The effluent stands in large tanks so the faecal / organic solids sink and are removed as sludge **1** This removes over 95% of the organic matter from the fluid so the risk of deoxygenation of the river is greatly reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water	2
03.3	**1 mark each** 1 mark for each of the 3 points below. **1** Aerobic bacteria digest / break down the remaining dissolved and suspended organic matter **1** Air (or pure oxygen) is mixed in by paddle wheels or air stones so oxygen shortage does not limit the aerobic process **1** The suspended bacteria collected in the secondary sedimentation tank are the 'activated sludge' – returning them maintains a large bacterial population so digestion is rapid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.4	**1 mark each** 1 mark for each of the 3 points below. **1** Effluent from secondary treatment may still contain dissolved inorganic nutrients (phosphates, nitrates) that cause eutrophication, and some bacteria **1** Needed where the discharge site is ecologically sensitive or used by many people (bathing water, drinking water abstraction) **1** Phosphate stripping with iron(III) sulfate – micro-strainers (very fine sieves) then UV light or chlorine to kill remaining bacteria **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 269-272; AQA spec 3.4.3.2.9.	
04.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Sludge from primary sedimentation is digested by anaerobic microbes in a warm tank for about four weeks **1** Kills most pathogens **1** Reduces odour and volume of the sludge **1** Produces biogas (mainly methane with some CO2) used as fuel to heat the digester / generate electricity **1** Digested sludge has a better C:N ratio and can be used as agricultural fertiliser **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
04.2	**1 mark each** 1 mark for each of the 4 points below. **1** Heavy metals from urban wastes / road runoff may be present and absorbed by crops - reduced by adding lime to lower metal solubility **1** Remaining pathogens - destroyed by heat treatment of the sludge before use **1 Timing:** spreading in the growing season covers crops – in winter vehicles compact wet soils - apply at suitable times / when soils are dry **1** Odour / public objection - digestion reduces odour **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Foul waste is a small volume needing full treatment so smaller, cheaper treatment works can be built – surface water needs little treatment **1** The treatment works is not affected by weather-related fluctuations in flow (storms) so storm overflows of untreated sewage are avoided **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 270-273; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** Spoil contains residual sulfide ores (eg iron pyrites, zinc sulfide) **1** These react with oxygen in the air spaces between the waste particles (oxidation) **1** Producing metal oxides and oxides of sulfur which dissolve in drainage water to form sulfuric acid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** The low pH (3.4) directly damages exposed cells / denatures proteins and dissolves calcium-based exoskeletons **1** Low pH increases the solubility of zinc (0.02 to 6.5 mg dm^{-3}) which is toxic (inhibits enzymes) **1** Only a few tolerant taxa survive – the sensitive mayfly larvae (indicator of clean water) are eliminated – dissolved oxygen is not the limiting factor here **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Dilution by tributaries / more water **1** Neutralisation by alkaline rocks (limestone) in the stream bed – zinc precipitates / adsorbs onto sediments as pH rises **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Collect the drainage water and pass it through crushed limestone / add lime to neutralise the acid **1** Higher pH also reduces the solubility of toxic metals so they precipitate and can be removed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9-3.4.3.2.10.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Many different organic substances may be present, each with a different potential to cause deoxygenation, so it is difficult to measure or forecast them individually **1** BOD measures the effect (amount of oxygen used by microbes decomposing the organic matter) under standardised conditions (eg 5 days, 20 °C, dark) **1** It gives a single measure directly related to the risk of deoxygenation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biochemical Oxygen Demand** → the oxygen used by aerobic decomposers in a water sample • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'a high BOD means the water contains a lot of oxygen' - a high BOD means a lot of oxygen will be removed	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** E. coli is a common gut bacterium so its presence confirms faecal contamination and its number indicates how serious the contamination is **1** Pathogens are not always present so their absence would not prove the water is uncontaminated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Different species have different sensitivities to pollution **1** Easy to identify **1** Easy to find / normally present / usually common **1** Widely distributed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** assesses the effect of pollution over a period of time (recent pollution events are detected even if the pollutant has gone) – no expensive equipment needed **1 Disadvantage:** identification needs skill – does not identify which pollutant is present so further chemical tests are required – species may be absent for other reasons **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 273-275; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Organic nutrients (sewage, manure, silage effluent, paper mill / tannery / food processing effluent) are food for aerobic microbes whose respiration deoxygenates the water, killing fish and invertebrates – bacterial growths cover the bed, blocking light • Pathogens in sewage (cholera, typhoid, dysentery) spread via contaminated water – decay of organic matter releases inorganic nutrients • **Inorganic nutrients:** nitrates (fertiliser / manure runoff, leaching) and phosphates (sewage effluent, detergents, eroded soil) cause cultural eutrophication: algal blooms, shading of macrophytes, cyanobacterial toxins, night-time and end-of-summer deoxygenation • **Human health:** nitrites and methaemoglobinaemia – possible carcinogen via nitrosamines • **Control of organic wastes:** sewage treatment (pre-treatment, primary sedimentation removing >95% organic matter, aeration / activated sludge or trickling filters, anaerobic digestion of sludge with biogas) • **Control of nutrients:** tertiary phosphate stripping with iron(III) sulfate, denitrification in filter beds, micro-strainers / UV – Nitrate Vulnerable Zones, fertiliser timing / amount, low-solubility fertilisers, legumes, manure management – dredging sediments • **Evaluation:** sewage treatment has virtually eliminated deoxygenation and waterborne disease in developed countries but storm overflows and combined sewers still discharge untreated sewage – tertiary treatment is expensive so only used at sensitive sites – diffuse agricultural nitrate is hard to control and aquifers respond slowly – phosphate legacy in sediments • Judgement on point-source (sewage) control being more successful than diffuse-source (agriculture) control	9

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Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **eutrophication** → nutrient enrichment of water leading to algal growth • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (9 marks)	
	Organic wastes - sewage, manure, silage effluent, paper-mill and food-processing effluent - are food for aerobic bacteria, whose respiration strips dissolved oxygen from the water so fish and invertebrates suffocate and bacterial growths smother the bed; sewage also carries pathogens such as cholera and typhoid. Inorganic nutrients act differently: nitrate leached from fertiliser and manure and phosphate from sewage, detergents and eroded soil cause cultural eutrophication, in which algal blooms shade macrophytes, cyanobacteria release toxins and night-time respiration and decay deoxygenate the water. Nitrate in drinking water can cause methaemoglobinaemia and may form carcinogenic nitrosamines.	
	Point sources are controlled by sewage treatment: screening and grit removal, primary sedimentation removing most organic solids, aeration with activated sludge or trickling filters, and anaerobic digestion of sludge yielding biogas. Nutrients need tertiary treatment - phosphate stripping with iron(III) sulfate, denitrifying filter beds, micro-strainers and UV. Diffuse agricultural sources are managed by Nitrate Vulnerable Zones, limits on the timing and amount of fertiliser, low-solubility fertilisers, legumes, manure management and buffer strips.	
	Sewage treatment has virtually eliminated river deoxygenation and waterborne disease in developed countries, but storm overflows on combined sewers still discharge untreated sewage, and tertiary treatment is expensive so it is used only where eutrophication is a risk. Diffuse nitrate from farmland is much harder to control because it comes from thousands of sources and depends on farmers' behaviour and the weather. Control of point sources has therefore been far more successful than control of diffuse ones.	
	Weaker answer - Level 1 (3 marks)	
	Sewage in rivers uses up the oxygen because bacteria break it down, so fish die. Fertilisers wash into rivers and cause eutrophication, where algae grow and block the light and then the water runs out of oxygen. Sewage is treated in sewage works to clean it before it goes into rivers. Farmers should use less fertiliser. Sewage treatment works well but fertiliser is still a problem.	
	What makes the difference	
	• Level 1 has the outline of both mechanisms but Level 3 explains them precisely (aerobic respiration, shading, night-time respiration, pathogens, health effects) and names the pollutant sources.	
	• Level 3 describes the stages of sewage treatment and the specific nutrient controls; Level 1 says 'sewage works' and 'less fertiliser'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation gives reasons (storm overflows, cost of tertiary treatment, diffuse sources) for its point-vs-diffuse judgement; Level 1 asserts it.	
07.2	1 mark for the point below. **1** Algae / cyanobacteria absorb nutrients directly from the water (not the sediments) so they use dissolved nutrients more easily and grow rapidly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-272; AQA spec 3.4.3.2.9.	

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